



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

DSA0613-Data Handling and Visualization for Data Analytics

3D Geospatial Visualization for Urban and Environmental Data

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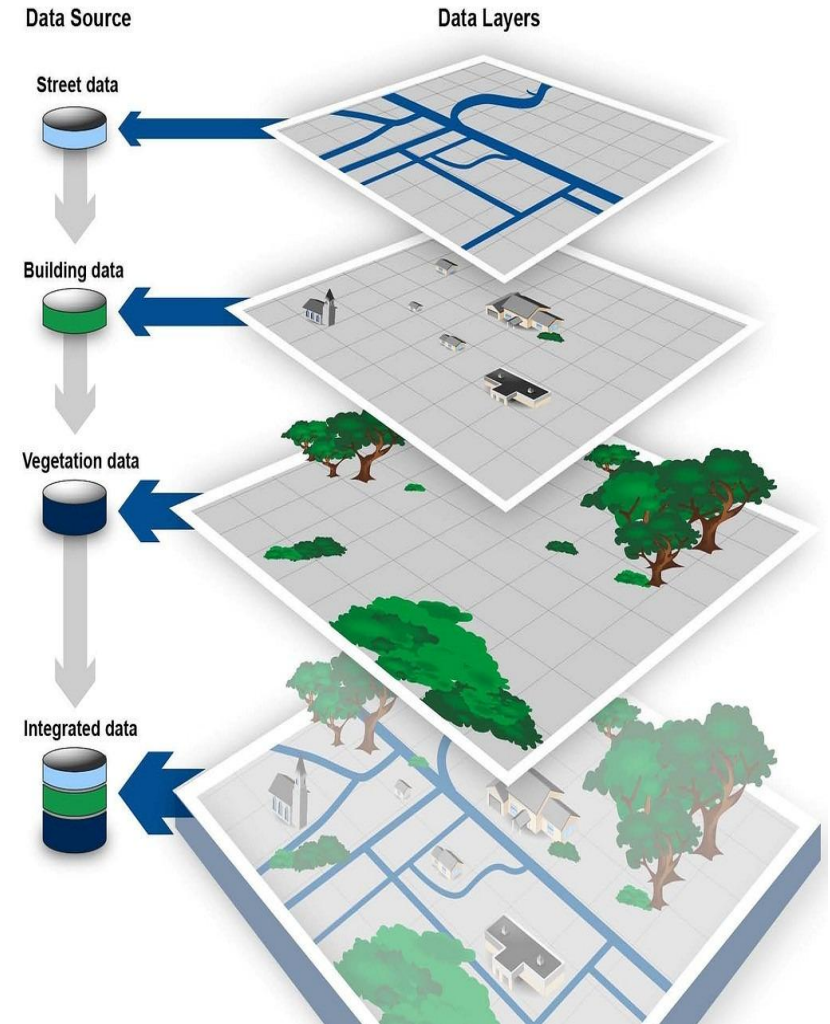
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INTRODUCTION

- 3D geospatial visualization represents spatial data in a realistic three-dimensional form.
- It helps in better understanding of urban and environmental structures.
- Traditional 2D maps have limitations in showing height and depth information.
- Integrating GIS and remote sensing improves spatial data analysis.
- This approach supports effective urban planning and environmental management.



ABSTRACT

- Represents urban and environmental data using realistic 3D geospatial models.
- Helps analyze complex spatial patterns like land use and urban growth.
- Integrates GIS, remote sensing, and real-time data sources.
- Supports effective decision-making in urban planning and environmental management.
- Improves data understanding, accuracy, and visual clarity.



PROBLEM STATEMENT

- Urban and environmental data are often **not integrated** for combined analysis.
- Environmental conditions vary across locations and are **hard to visualize spatially**.
- Most tools rely on **static 2D maps**, limiting spatial understanding.
- Integrating **real-time environmental data** with urban context is challenging.
- There is a need for an **interactive 3D geospatial visualization system** for better analysis and decisions.

OBJECTIVES

- To collect and preprocess **urban and environmental geospatial data**.
- To develop an **interactive 2D and 3D visualization system**.
- To visualize **urban structures and environmental parameters** effectively.
- To enable **comparative analysis** across multiple geographic locations.
- To support **decision-making** through intuitive visual insights.

DATASET

- Urban Reference Dataset:**

Cities and towns with latitude and longitude.

Source: Compiled reference dataset (validated using OpenStreetMap coordinates).

- Environmental Dataset (Real-Time):**

Air Quality, Temperature, Humidity, Wind Speed.

Source: Open-Meteo public APIs.

- Visualization Data:**

2D maps and 3D interactive models.

Source: Leaflet and Plotly libraries in R.

METHODOLOGY

Data Collection:

Urban reference data and real-time environmental data are gathered from datasets and public APIs.

•Data Preprocessing:

Coordinates are validated and environmental values are prepared for visualization.

•User Interaction:

Locations are selected through an interactive map interface.

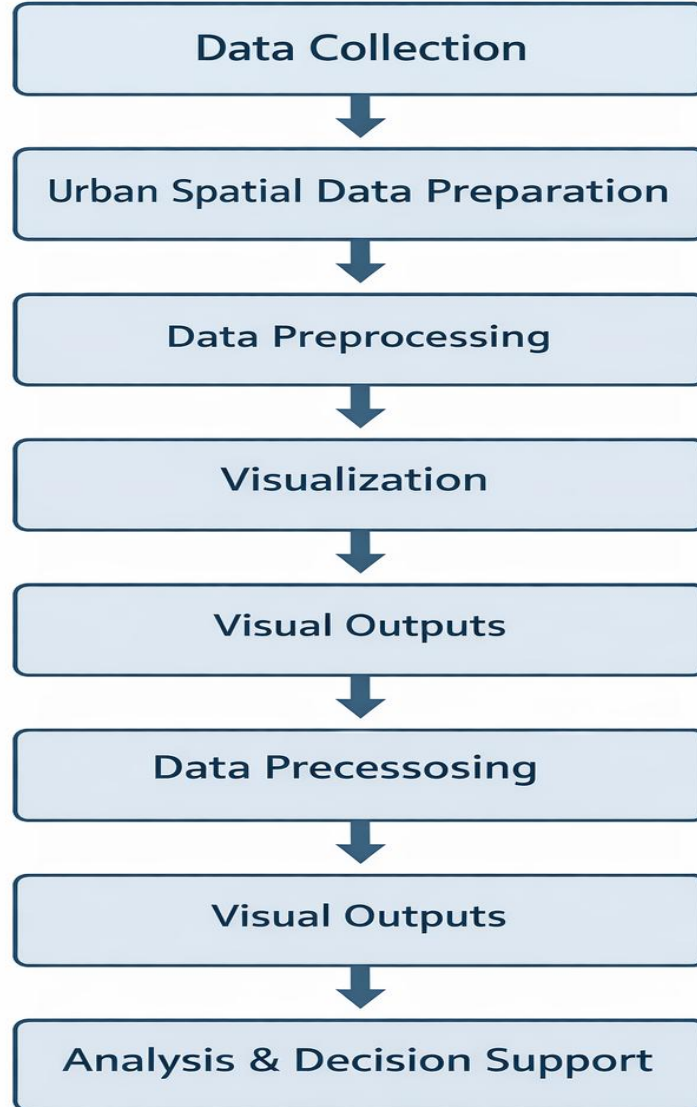
•3D Visualization:

Urban and environmental data are displayed using interactive 3D models.

•Analysis and Insights:

Visual comparisons are used to support decision-making.

ARCHITECTURE



MODULE 1

Collect and Preprocess Geospatial Urban and Environmental Data

- Collects **urban reference data**
- Collects **real-time environmental data** (AQI, Temperature, Humidity, Wind Speed)
- Performs preprocessing:
 - coordinate validation
 - data cleaning and scaling

Technologies involved:

- Open- Meteo APIs
- Excel dataset
- R (data preprocessing functions)

MODULE 2

Build Interactive 3D Models and Visualizations

- Interactive 2D map for selecting geographic locations
- Real-time update of visualizations based on user interaction
- Abstract 3D urban modeling using:
 - Grey 3D building blocks
 - Green cover patches
 - Water body patches
- Layer-based visualization for better clarity and comparison
- Zooming, rotation, and panning of 3D models

➤ **Libraries Used:**

- **Shiny** – To build the interactive web-based application
- **Leaflet** – For 2D geospatial mapping and location selection
- **Plotly** – For interactive 3D visualizations and models
- **ggplot2** – For generating supporting analytical graphs
- **jsonlite** – To parse real-time data from web APIs
- **readxl** – To load and process urban reference datasets

MODULE 3

Analyze Data and Generate Decision-Support Insights

- Aggregation of environmental data from multiple user-selected locations
- Visual comparison of AQI levels to assess pollution severity
- Evaluation of temperature and wind speed trends across regions
- Assessment of humidity distribution for climatic understanding
- Identification of spatial differences in environmental conditions

➤ **Analytical Visualizations Used:**

- **Bar Charts** – AQI and Humidity comparison
- **Dot Plots** – Temperature and Wind Speed variation

IMPLIMENTATION:

```
install.packages(c("shiny", "plotly", "dplyr"))
install.packages("leaflet")
library(shiny)
library(plotly)
library(dplyr)
library(leaflet)

# ----- REAL-TIME URBAN + ENVIRONMENT DATA -----
generate_realtime_data <- function() {
  data.frame(
    Building = c("Apartment", "Factory", "Office"),
    Longitude = c(78.4867, 78.5000, 78.4900),
    Latitude = c(17.3850, 17.3900, 17.3880),
    Height = c(
      sample(40:80, 1), # Apartment height
      sample(60:120, 1), # Factory height
      sample(30:70, 1) # Office height
    ),
    Pollution = c(
      sample(20:60, 1),
      sample(90:150, 1),
      sample(40:80, 1)
    )
  )
}
```

```
# ----- UI -----
ui <- fluidPage(
  titlePanel("Real-Time 3D Urban Buildings & Environmental Data"),

  fluidRow(
    column(6, leafletOutput("map", height = "500px")),
    column(6, plotlyOutput("plot3d", height = "500px"))
  )
)

# ----- SERVER -----
server <- function(input, output, session) {

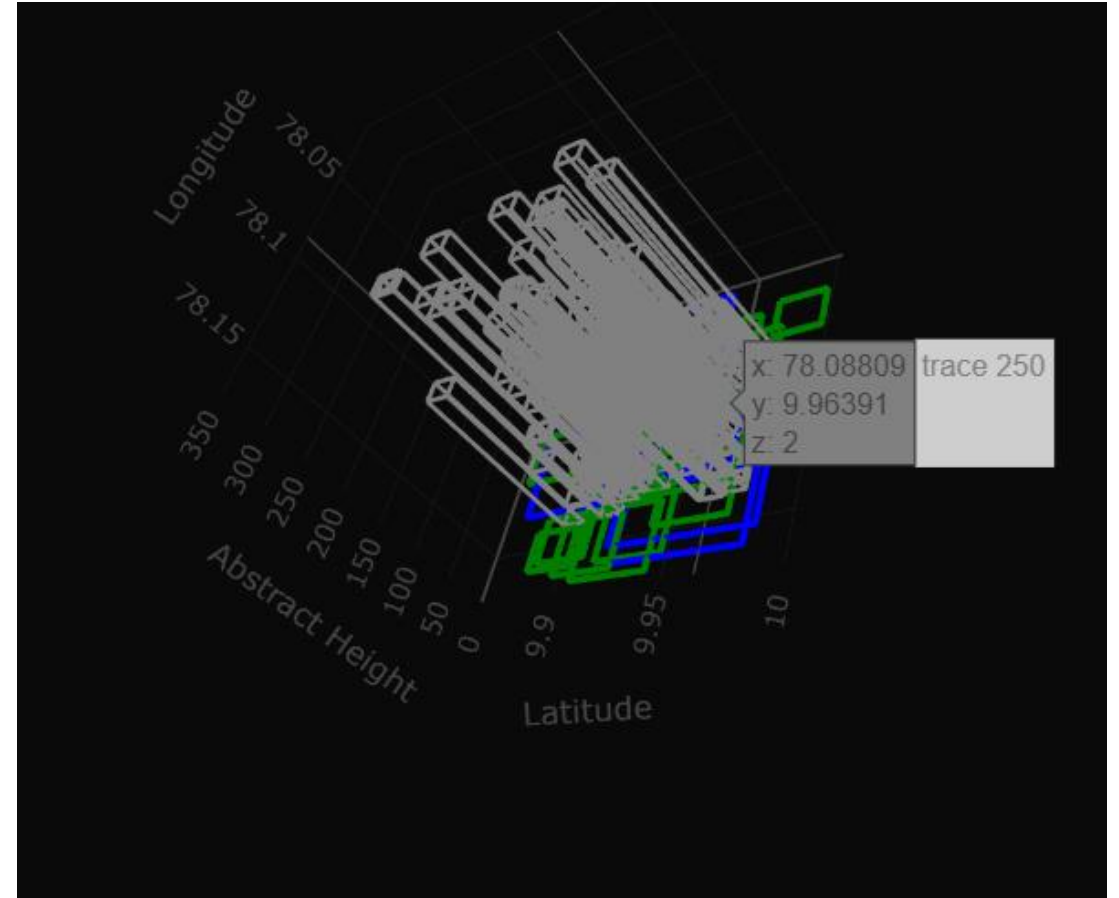
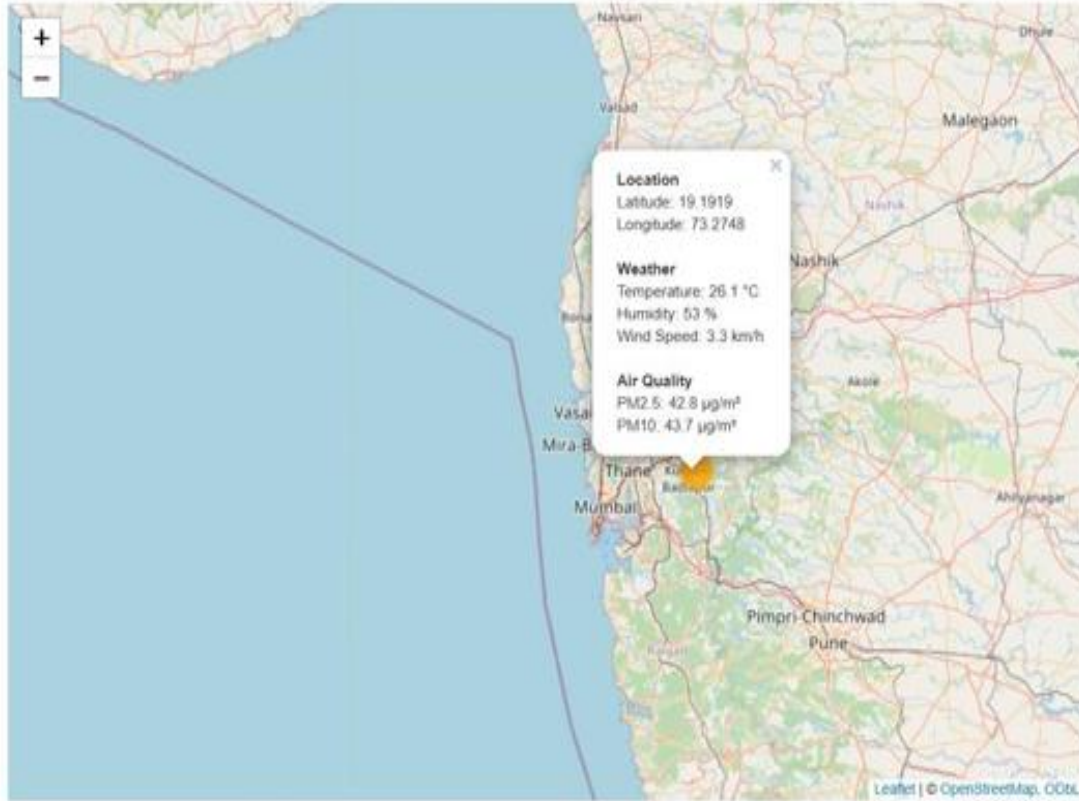
  realtime_data <- reactiveVal(generate_realtime_data())

  # Update every 5 seconds
  observe({
    invalidateLater(5000, session)
    realtime_data(generate_realtime_data())
  })

  # ---- MAP (2D) ----
  output$map <- renderLeaflet({
    leaflet() %>%
```

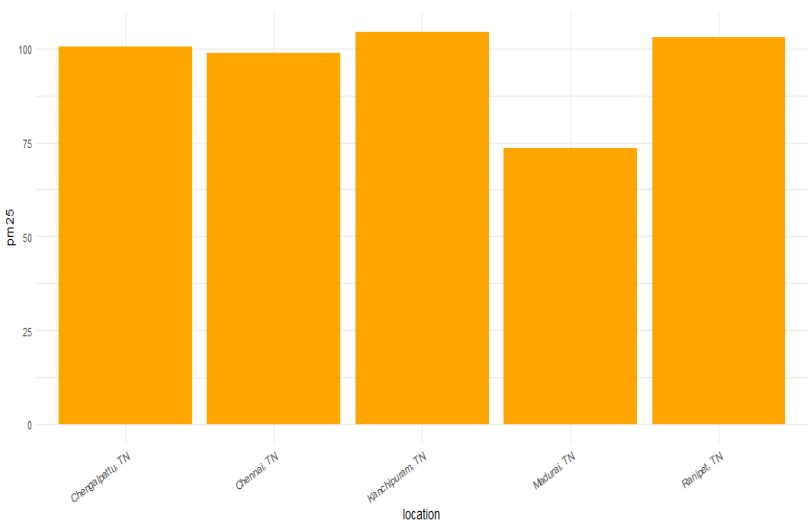
OUTPUT:

Global Urban & Environmental Data Viewer

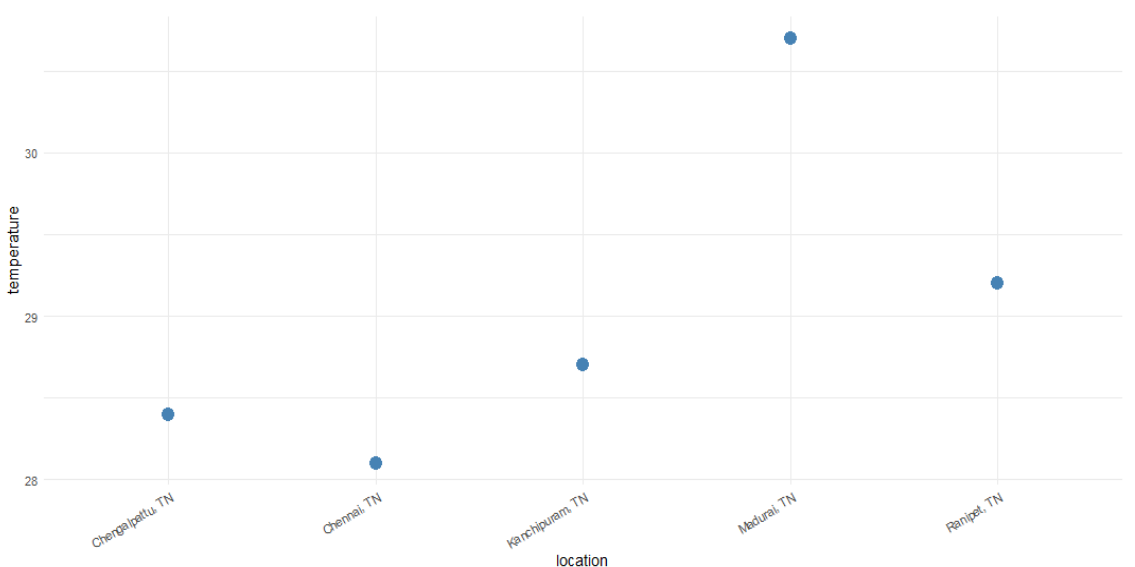


OUTPUT:

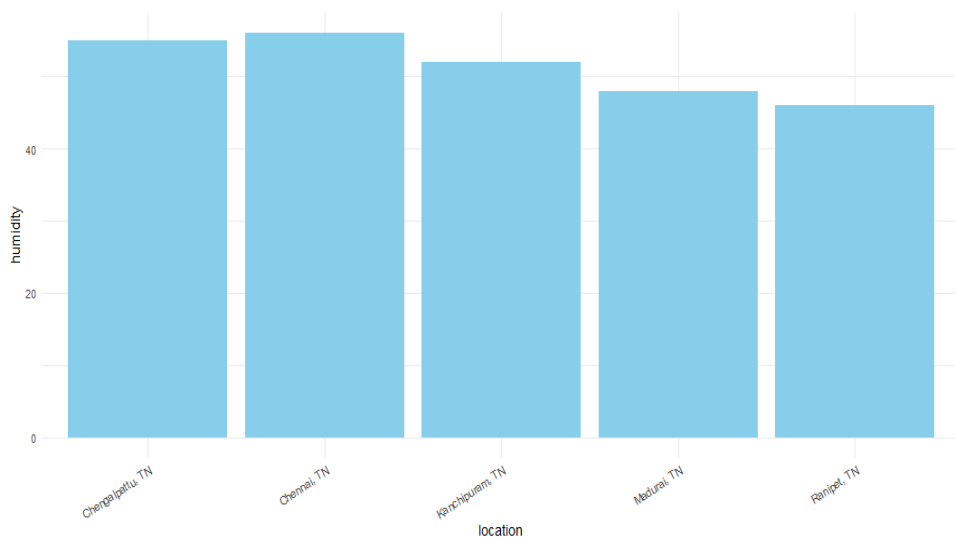
AQI (PM2.5)



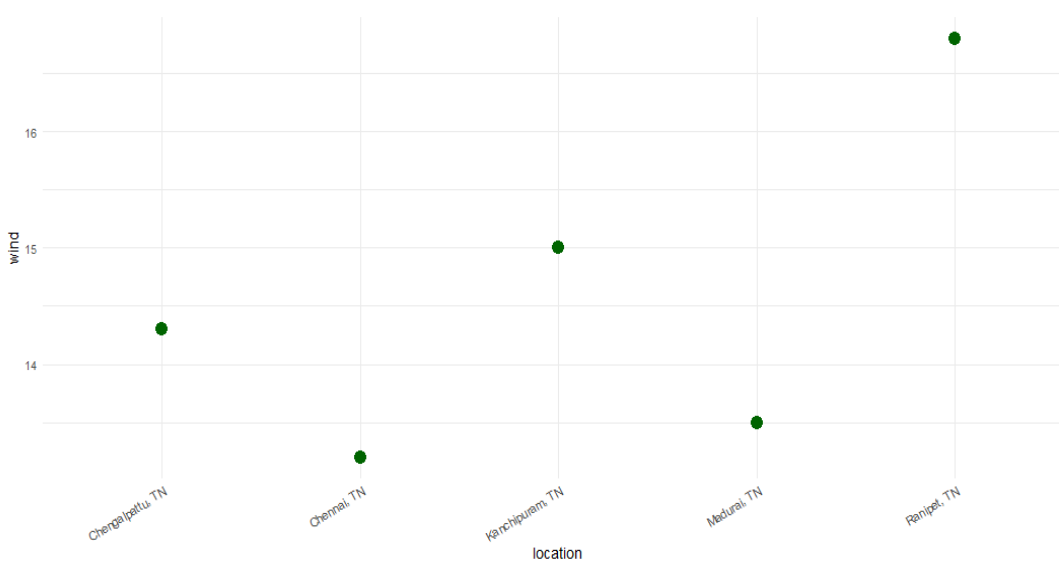
Temperature



Humidity



Wind Speed



CONCLUSION

- 3D geospatial visualization effectively represents complex urban and environmental data.
- It improves understanding of spatial relationships compared to traditional 2D maps.
- Integration of multiple geospatial data sources enhances analysis accuracy.
- The system supports better urban planning and environmental decision-making.
- Overall, 3D visualization provides clearer insights and more informed solutions.

REFERENCES

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- Liu, X., & Zhang, Y. (2023). 3D geospatial visualization techniques in urban planning. *Computers, Environment and Urban Systems*.
- La Guardia, M. et al. (2024). Real-time 3D geospatial data integration for smart cities. *ISPRS Archives*.
- Somanath, S. et al. (2024). Urban digital twins using 3D geospatial data. *Remote Sensing*.
- Lei, B., & Biljecki, F. (2025). Advances in open 3D city models. *ISPRS Annals*.

FUTURE SCOPE

- Integration of real-time IoT and sensor data for live urban monitoring.
- Use of AI and machine learning to predict urban growth and environmental changes.
- Development of digital twins for smart city planning and simulation.
- Enhanced VR and AR-based 3D visualization for immersive analysis.
- Cloud-based platforms for large-scale data sharing and collaborative decision-making.

**THANK
YOU**